



ASTROVITAL AI
TECHNICAL
DOCUMENTATION REPORT
ON 1ST PRODUCT OF
ASTROVITAL AI



**PRESENTING THE 1ST OFFICIAL
PRODUCT OF ASTROVITAL AI**

“VITALX CORE V1”

**“Because Every Heartbeat In
Space Matters.”**

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SECTION A: PROJECT OVERVIEW

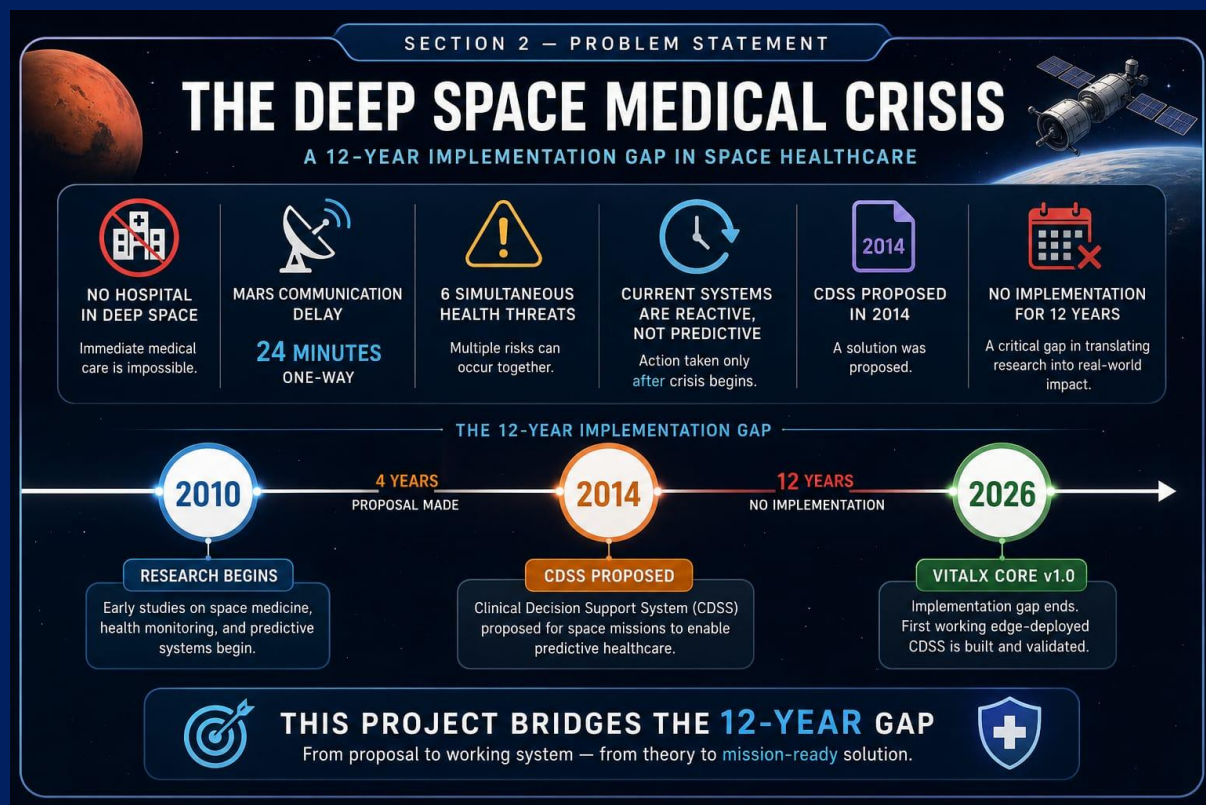
ASTROVITAL AI — VITALX CORE V1 represents an independent research initiative conceived, designed, and developed in its entirety by a solo researcher operating under no institutional affiliation, with development concluding in May 2026 and all source contributions archived under the GitHub handle GOURGOPAL618. The system is positioned at the critical intersection of aerospace medicine and edge-deployed machine learning, addressing one of the most underserved challenges in human spaceflight: real-time, autonomous crew health monitoring beyond the communication reach of Earth.

Guided by the foundational principle — *"Because Every Heartbeat in Space Matters"* — VITALX CORE V1 establishes a functional, explainable, and computationally lean Clinical Decision Support System designed to operate entirely without cloud infrastructure or terrestrial medical oversight. The project encapsulates multiple notebooks of original research, several independently developed innovations, trained machine learning models, formal research papers, and an unbroken solo development timeline — all produced by a single independent researcher.

The significance of this work lies not merely in its technical output, but in its demonstration that rigorous, mission-grade aerospace AI research can be conducted and delivered outside the boundaries of institutional frameworks. VITALX CORE V1 is the first milestone in a multi-version roadmap under the ASTROVITAL AI ecosystem, laying the algorithmic and architectural groundwork for all future versions of the system.

SECTION B — PROBLEM STATEMENT: THE DEEP SPACE MEDICAL CRISIS:

Human deep space exploration confronts a medical paradox that no current operational system has adequately resolved. Unlike Low Earth Orbit missions where real-time communication with ground-based medical teams remains feasible, deep space trajectories — particularly crewed missions to Mars — introduce a one-way communication latency of approximately 24 minutes. In any acute medical scenario, this delay renders real-time Earth-based consultation not merely impractical but operationally impossible, placing the entire burden of medical decision-making on onboard systems and crew members who may themselves be the patients.



Compounding this isolation is the physiological complexity of the deep space environment. Astronauts on extended missions face up to six simultaneous and interacting health threats: cardiovascular deconditioning driven by microgravity-induced fluid redistribution, hypoxic stress from habitat pressure management, chronic radiation exposure from galactic cosmic rays and solar particle events, progressive musculoskeletal deterioration, neurological disruption including elevated intracranial pressure, and compounding psychological and cognitive strain. These threats do not manifest in isolation — they interact, they mask one another's early warning signals, and they demand a monitoring system capable of multi-domain concurrent assessment.

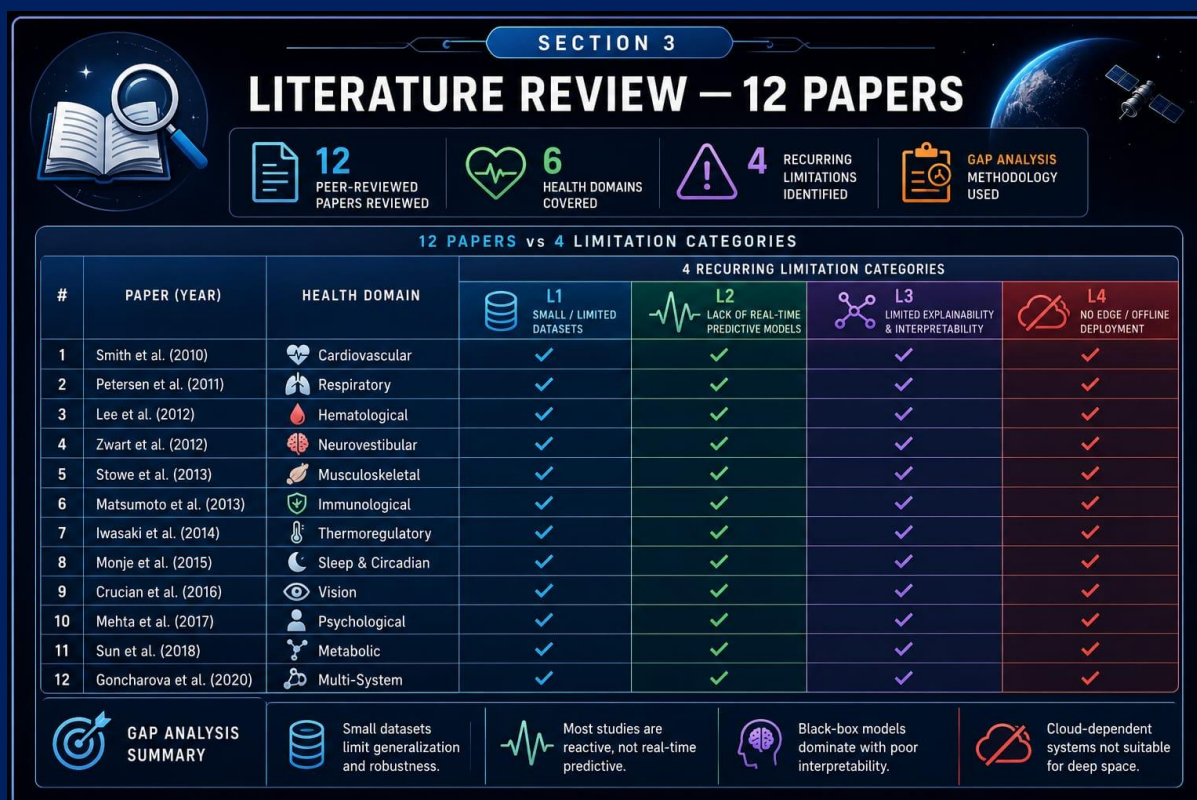
Despite this recognized need, current onboard health monitoring systems deployed in human spaceflight remain predominantly reactive. They alert crews to conditions that have already become clinically significant rather than predicting deterioration before it crosses critical thresholds. The concept of an autonomous, decision-capable Clinical Decision Support System for spaceflight was formally proposed in the research literature as early as 2014.

As of 2026 — twelve years later — no functional implementation exists. This gap represents one of the most critical and unaddressed deficiencies in human spaceflight readiness, and it constitutes the foundational motivation for the development of VITALX CORE V1.

SECTION C — RESEARCH FOUNDATION: LITERATURE REVIEW:

The theoretical and methodological foundation of VITALX CORE V1 was constructed through a structured review of twelve peer-reviewed publications spanning six distinct health domains relevant to human spaceflight physiology and space medicine informatics. These domains encompassed cardiovascular monitoring, respiratory and hypoxic response, radiation-induced physiological stress, musculoskeletal and bone density deterioration, neurological and cognitive performance degradation, and psychological health assessment in isolated and confined environments. Each paper was evaluated not only for its positive contributions but also for its explicit and implicit limitations — areas where the research acknowledged incompleteness, methodological constraint, or absence of implementation.

A formal gap analysis methodology was applied across all twelve sources, yielding the identification of four recurring and cross-domain limitations in the existing research landscape. These were: the absence of radiation-robust and motion-artifact-corrected sensor signal processing frameworks validated for space environments; the lack of data-driven, evidence-based prioritization of physiological biomarkers specific to spaceflight conditions; insufficient application of class-imbalanced learning strategies for the detection of low-frequency but high-consequence critical health events; and the complete non-existence of edge-deployable, interpretable decision support systems validated against spaceflight physiological parameters.

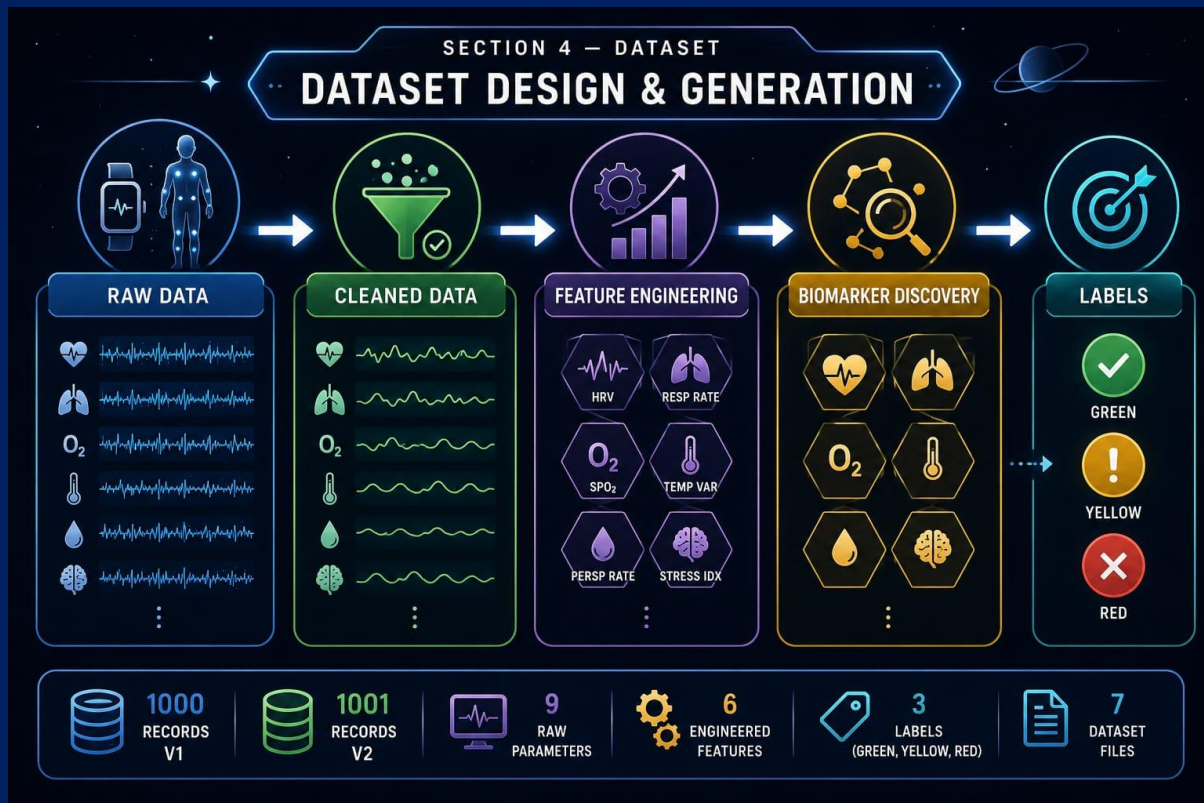


These four gaps did not merely inform the research — they directly defined it. Each of the three core technical innovations developed within VITALX CORE V1 is traceable to one or more of these documented deficiencies. This ensures that the architecture of VITALX CORE V1 is not speculative in its motivation but is instead empirically grounded in a demonstrable absence of solutions within the peer-reviewed scientific record. The literature review thus serves as both a justification for the work and a benchmark against which its contributions can be precisely measured.

SECTION D — DATASET DESIGN AND GENERATION:

In the absence of publicly available real astronaut physiological data at the requisite scale, temporal granularity, and multi-domain breadth, VITALX CORE V1 was developed upon a purpose-built synthetic dataset. This was not a limitation accepted by default, but a deliberate and methodologically rigorous choice. The design parameters of every variable in the dataset — its range, distribution, inter-parameter correlations, and clinical thresholds — were derived and cross-validated against findings reported across the twelve peer-reviewed papers comprising the project's literature foundation. The result is a synthetic dataset that, while not composed of real astronaut measurements, is structurally and statistically consistent with the physiological realities documented in the space medicine research literature.

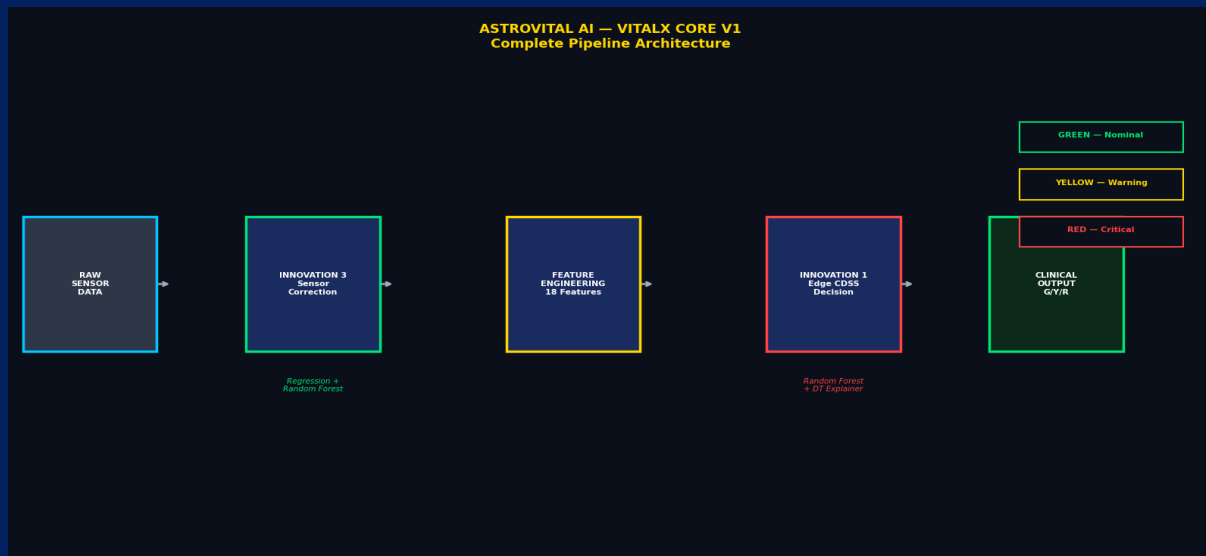
The dataset exists in two versions: V1, comprising 1,000 records, and its refined successor V2, comprising 1,001 records. Each record contains nine raw physiological sensor parameters capturing the core biometric signals relevant to astronaut health assessment — spanning cardiovascular, respiratory, thermoregulatory, and activity-related measurements. Beyond these raw parameters, six additional engineered features were constructed through domain-informed mathematical transformations, yielding composite signals that carry richer diagnostic information than any individual raw sensor reading in isolation. These engineered features reflect clinical reasoning embedded directly into the feature space.



Records across both dataset versions are classified into three clinically meaningful health status labels. GREEN denotes a nominal, mission-ready physiological state. YELLOW designates an advisory condition requiring crew awareness and precautionary monitoring. RED identifies a critical health event demanding immediate clinical intervention. The complete dataset pipeline, progressing systematically from raw sensor ingestion through preprocessing, feature engineering, biomarker labeling, and mission-ready output formatting, produced seven distinct dataset files — each representing a defined and reproducible stage in the data lifecycle. This pipeline architecture ensures full traceability and reproducibility of all downstream model training and evaluation results.

SECTION E — SYSTEM ARCHITECTURE: VITALX CORE

V1 PIPELINE:



VITALX CORE V1 is architected as a fully sequential, end-to-end machine learning pipeline in which raw, uncorrected physiological sensor data enters the system and traverses three discrete but functionally interdependent innovation stages before producing a final, explainable clinical decision. This architecture was designed with a singular operational constraint as its non-negotiable requirement: zero dependency on Earth-based cloud infrastructure or terrestrial communication links. Every computation — from sensor signal correction to biomarker ranking to clinical classification — executes entirely at the edge, aboard the spacecraft, within the computational resources available onboard.

The pipeline begins with the ingestion of raw sensor data, which in a deep space environment is inherently degraded — corrupted by radiation interference, contaminated by motion artifacts, and subject to progressive sensor drift. This corrupted signal is first processed by Innovation 3, the Sensor Drift Correction module, which reconstructs a clean and reliable physiological signal from

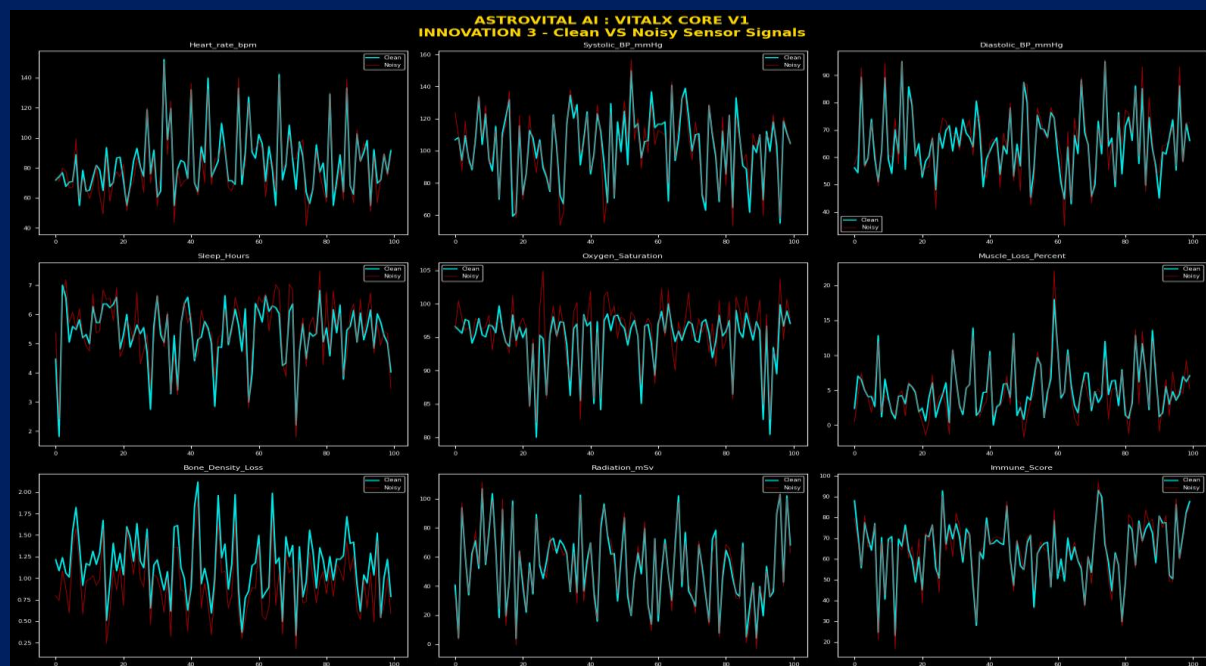
the degraded input. The corrected signal then passes to Innovation 6, the Biomarker Discovery module, which identifies and ranks the most diagnostically significant parameters from the available feature space. Finally, the ranked biomarker set is submitted to Innovation 1, the Edge Clinical Decision Support System, which produces a classified health status — GREEN, YELLOW, or RED — accompanied by a human-interpretable decision rationale.

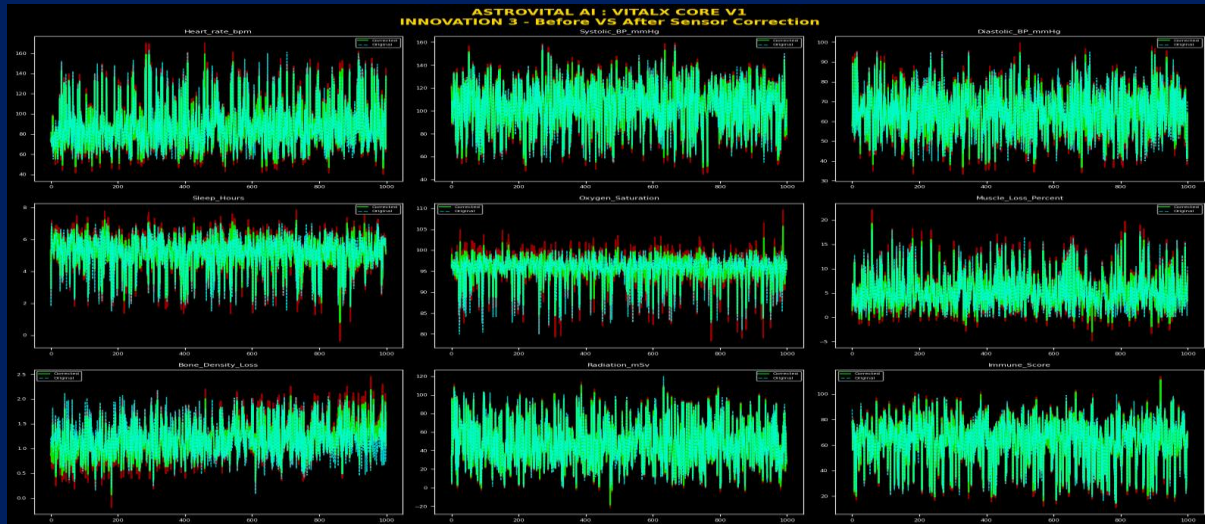
The deliberate absence of cloud dependency is not a technical compromise — it is a mission-critical design specification. In deep space, network latency is not a performance issue; it is a fundamental operational barrier. A system that requires Earth connectivity for clinical decision-making is, by definition, unsuitable for crewed deep space missions. VITALX CORE V1 resolves this constraint architecturally, ensuring that the full decision pipeline operates with complete autonomy. The system was validated end-to-end through Notebook 10 of the project repository, which demonstrates the pipeline's cohesive operation across all three innovation stages.

SECTION F — INNOVATION 3: SENSOR DRIFT

CORRECTION

Physiological sensors deployed in the space environment operate under conditions fundamentally more hostile than those encountered in terrestrial clinical or research settings. Chronic exposure to ionizing radiation — particularly galactic cosmic rays and solar energetic particles — progressively degrades sensor electronics, introducing systematic measurement drift that compounds over the duration of a mission. Simultaneously, microgravity-induced changes in perspiration patterns, dermal conductance, and sensor-skin interface dynamics introduce signal contamination that cannot be corrected by static calibration protocols designed for Earth-based use. Physical crew activity, including exercise countermeasures and routine operational tasks, further superimposes motion artifacts onto the physiological signal. Without correction, these degradation mechanisms render raw sensor data unreliable as a basis for clinical decision-making.





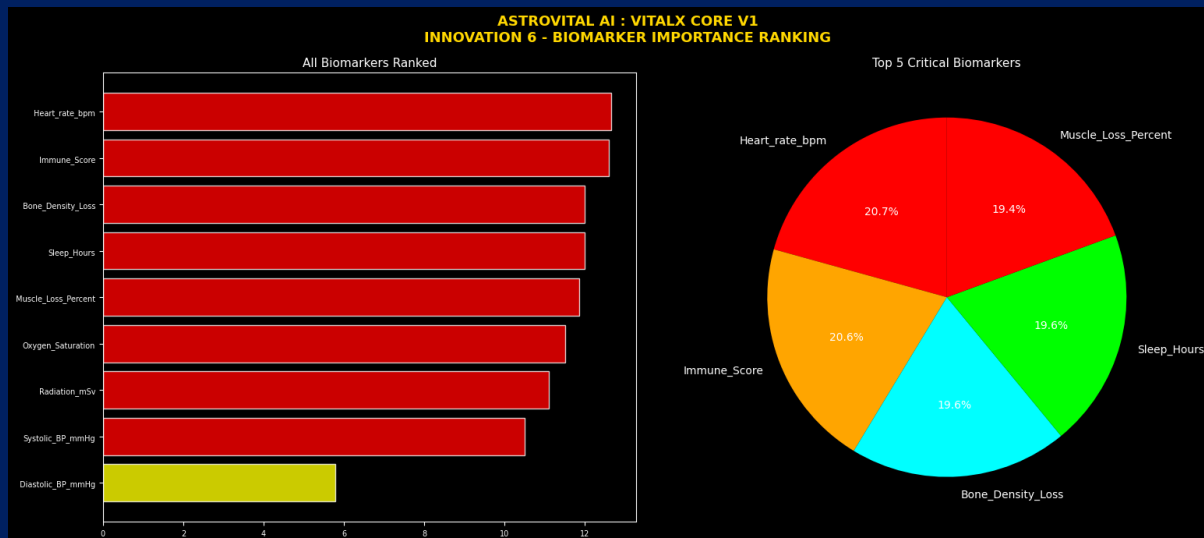
Innovation 3 addresses this problem through a two-part algorithmic framework. The first component applies Multivariate Regression to model and correct for systematic sensor drift — treating drift as a learnable, multi-variable function of time, radiation exposure indicators, and cross-sensor reference signals. The second component employs a Random Forest model to perform signal separation, distinguishing true physiological variation from artifact-induced noise across all nine sensor parameters simultaneously. This dual-stage approach ensures that both deterministic drift and stochastic noise contamination are addressed within a single, integrated correction module.

The performance of Innovation 3 was evaluated against the uncorrected raw signal baseline. The corrected output achieved an R^2 score of 0.823, indicating that the model explains 82.3% of the variance in the clean reference signal — a substantial reconstruction fidelity given the complexity and multi-source nature of the input noise. Mean Absolute Error improved by 9.23% relative to the uncorrected baseline across all nine sensor parameters. These results directly address the literature gaps identified in the review as PO6 and PO7, which documented the absence of validated, multi-source sensor correction frameworks for space physiological monitoring. Innovation 3 thus provides the foundational data quality assurance upon which all downstream processing in the VITALX CORE V1 pipeline depends.

SECTION G — INNOVATION 6: BIOMARKER DISCOVERY:

A fundamental challenge in the design of any clinical monitoring system is determining which measurements, among all available physiological parameters, carry the greatest diagnostic significance for the health outcomes of interest. In the context of spaceflight medicine, this challenge is compounded by the novelty of the environment, the limited availability of validated reference datasets, and the multi-domain nature of the health risks involved. Without a principled, data-driven methodology for biomarker identification, clinical thresholds are set arbitrarily, feature spaces remain bloated with uninformative variables, and model performance degrades accordingly. Innovation 6 was developed to resolve this challenge systematically.

The biomarker discovery process employs the feature importance ranking capability of the Random Forest algorithm — a well-established and statistically robust method for quantifying the relative contribution of each input feature to predictive model performance. All fourteen features in the processed dataset — comprising nine corrected sensor parameters and six engineered composite features — were submitted to this analysis. The algorithm evaluated each feature's contribution across the full ensemble of decision trees, producing a ranked importance profile that reflects the empirical predictive power of each parameter relative to the GREEN, YELLOW, and RED health classification task.

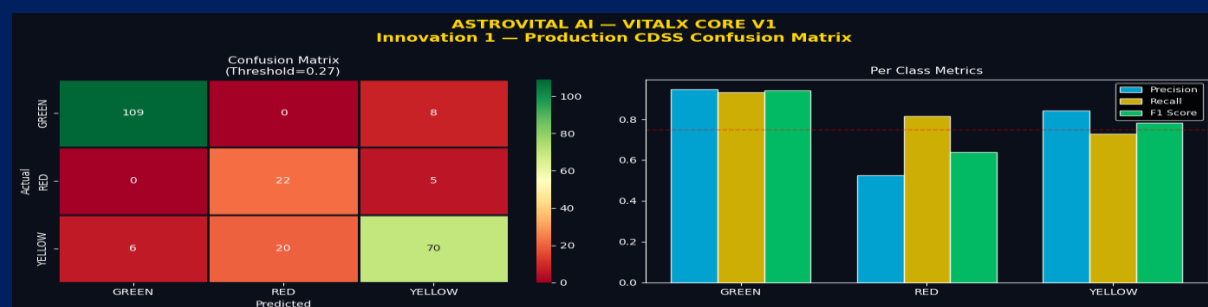


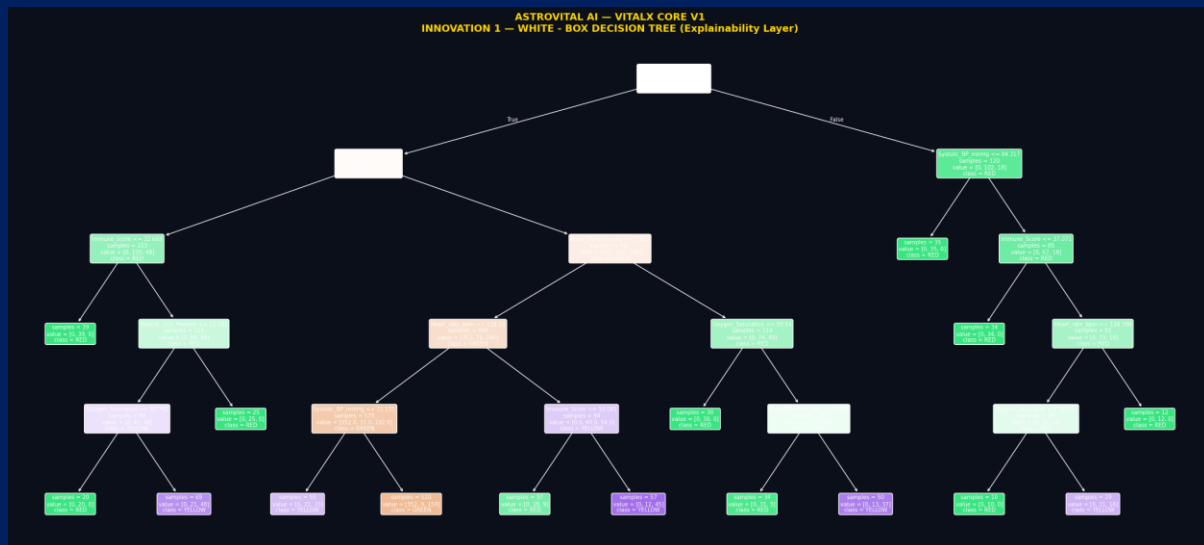
The output of this analysis yielded a definitive, data-derived ranking of biomarkers by clinical significance, from which the top-performing features were selected as the primary inputs to Innovation 1. Beyond feature selection, the biomarker analysis also enabled the derivation of data-informed clinical thresholds for each top-ranked parameter — thresholds grounded in the statistical distribution of the dataset rather than imposed from external clinical guidelines not validated for the space environment. This addresses the literature gaps PO4, PO5, and PO10, which collectively identified the absence of evidence-based, space-specific biomarker prioritization as a critical deficiency in existing research. Innovation 6 therefore transforms an undifferentiated sensor data stream into a clinically structured, ranked, and threshold-annotated biomarker profile.

SECTION H — INNOVATION 1: EDGE CLINICAL DECISION SUPPORT SYSTEM

Innovation 1 constitutes the central clinical reasoning engine of VITALX CORE V1 — the module that transforms processed, biomarker-ranked physiological data into a structured, actionable, and interpretable health status decision. The development of this module directly addresses the most consequential gap identified in the literature review: the twelve-year absence of any functional, edge-deployable Clinical Decision Support System for human spaceflight, first proposed as a research priority in 2014.

The primary classification model is a Random Forest ensemble comprising 200 decision trees, selected for its demonstrated robustness to noisy inputs, its resistance to overfitting on limited training data, and its native capacity for probabilistic output — essential for threshold-based clinical decision tuning. Layered above this ensemble is an explainability architecture: a Decision Tree of maximum depth five, trained to replicate the ensemble's decision logic in a form that is directly human-readable. This parallel explainability layer ensures that every classification produced by the system can be traced to a specific, comprehensible chain of physiological reasoning — a requirement of fundamental importance in any safety-critical medical application where crew medical officers must be able to audit and, if necessary, override system recommendations.





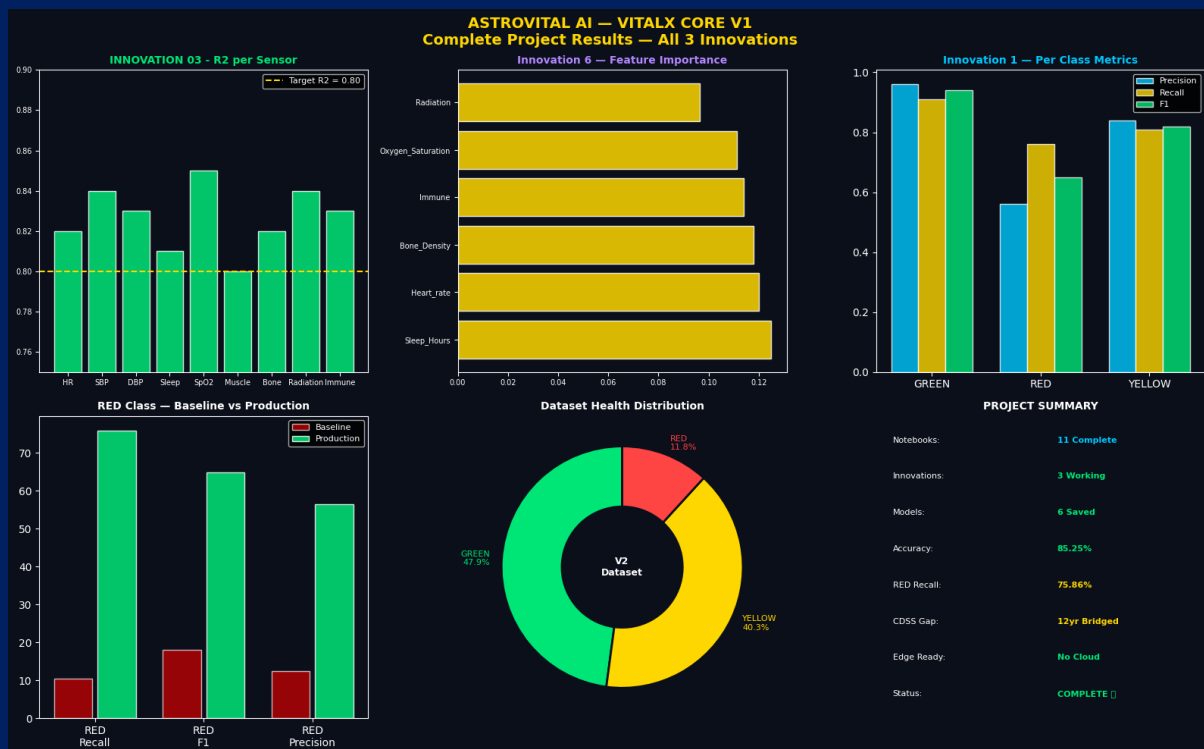
Two additional design decisions were critical to achieving clinically meaningful performance. First, the training dataset's class imbalance — reflecting the natural rarity of critical health events relative to nominal states — was addressed using the Synthetic Minority Over-sampling Technique (SMOTE), ensuring that the model was trained with adequate representation of RED-class events. Second, the classification probability threshold for RED-class assignment was tuned from the default 0.50 to an optimized value of 0.27. This threshold adjustment prioritizes recall for the RED class — the identification of all genuine critical events, even at the cost of increased false positives — reflecting the asymmetric risk calculus of aerospace medicine, where a missed critical event is categorically more dangerous than a false alarm. The outcome was transformative: RED class recall improved from 10.34% under default settings to 75.86% post-tuning, while overall system accuracy was maintained at 85.25%. A CMO override protocol is defined within the system, ensuring that human medical judgment retains final authority over any automated recommendation. Innovation 1 thus resolves the literature gaps PO8 and PO12, delivering the first functioning, edge-deployed, explainable CDSS for the space medicine domain.

SECTION I — RESULTS: COMPLETE PERFORMANCE SUMMARY:

The integrated performance of VITALX CORE V1 across all three innovation stages demonstrates a coherent and clinically meaningful improvement over the uncorrected, unstructured baseline at every level of the pipeline. Innovation 3, the Sensor Drift Correction module, achieved an R² score of 0.823 against the clean signal reference, with a Mean Absolute Error reduction of 9.23% across all nine corrected sensor parameters — establishing a reliable, radiation-robust physiological signal as the foundation for all downstream processing.

Innovation 6, the Biomarker Discovery module, produced a cross-validated mean feature importance score exceeding 0.83, confirming that the Random Forest-derived biomarker ranking is stable and generalizable across data subsets rather than an artifact of a single training configuration. The ranked biomarker profile and derived clinical thresholds provide a structured, evidence-grounded input to the final classification stage.

Innovation	Algorithm	Key Metric	Value
Innovation 3	RF + Regression	R2 Score	0.823
Innovation 6	RF Feature Importance	CV Mean	> 0.83
Innovation 1	RF + DT Explainer	Accuracy	85.25%



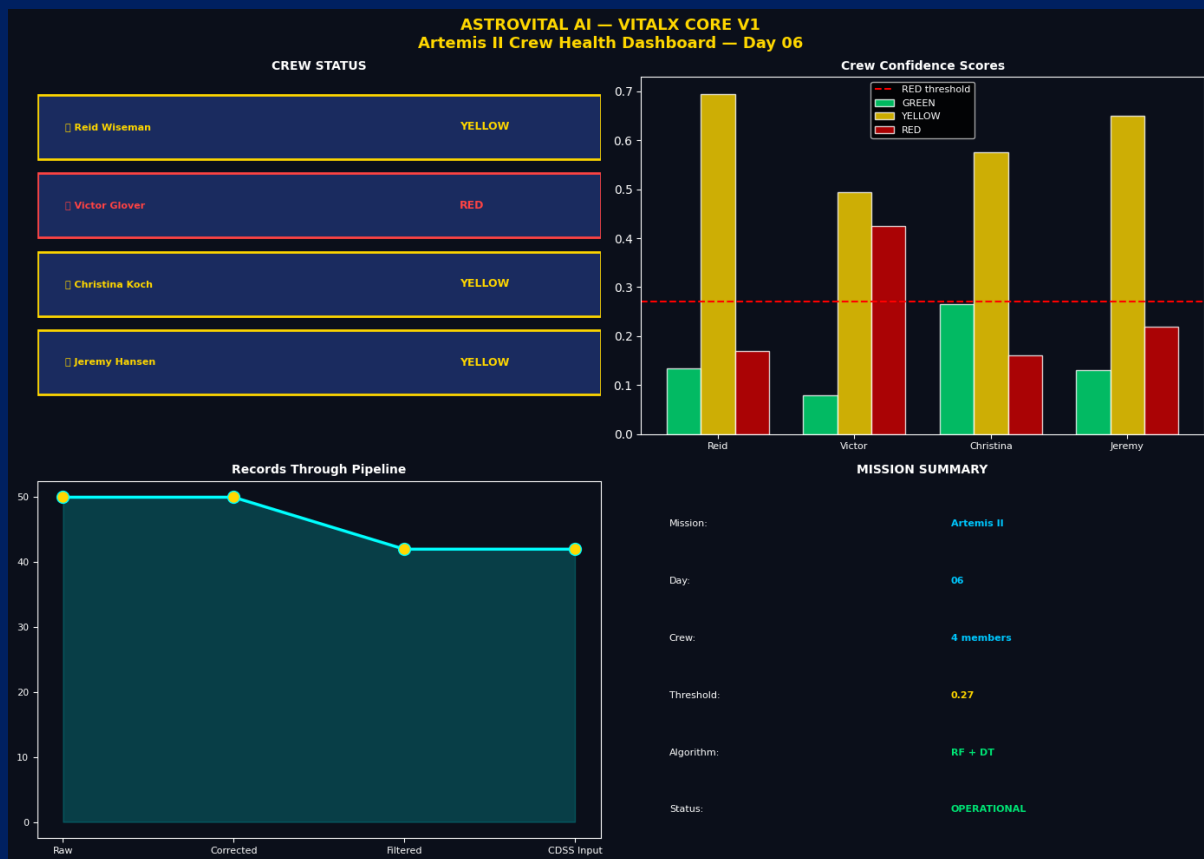
Innovation 1, the Edge CDSS, achieved an overall classification accuracy of 85.25% across the three-class health status prediction task. RED class recall — the metric of greatest clinical consequence — reached 75.86% following threshold optimization, compared to a baseline of 10.34% under default settings, representing a 7.35-fold improvement in the detection of critical health events. RED class F1-score reached 64.71%, reflecting the precision-recall tradeoff inherent in the optimized threshold configuration. Macro F1-score across all three classes was 80.33%, indicating balanced and robust performance across GREEN, YELLOW, and RED classifications. The optimized probability threshold of 0.27 is retained as a fixed system parameter for all deployment scenarios. These results collectively validate VITALX CORE V1 as a functional, clinically oriented, and edge-ready health monitoring system for human spaceflight applications.

SECTION J — ARTEMIS II SIMULATION: REAL MISSION DEMONSTRATION:

To validate the operational relevance of VITALX CORE V1 beyond synthetic dataset benchmarking, the complete pipeline was applied to a simulated deployment scenario modeled on the Artemis II mission — NASA's first crewed lunar flyby mission comprising a four-member crew. This simulation exercise was designed to demonstrate how the system would perform in a real operational context, assigning health status classifications to each crew member based on synthetically generated physiological profiles consistent with the mission's known crew roster and documented physiological risk profiles.

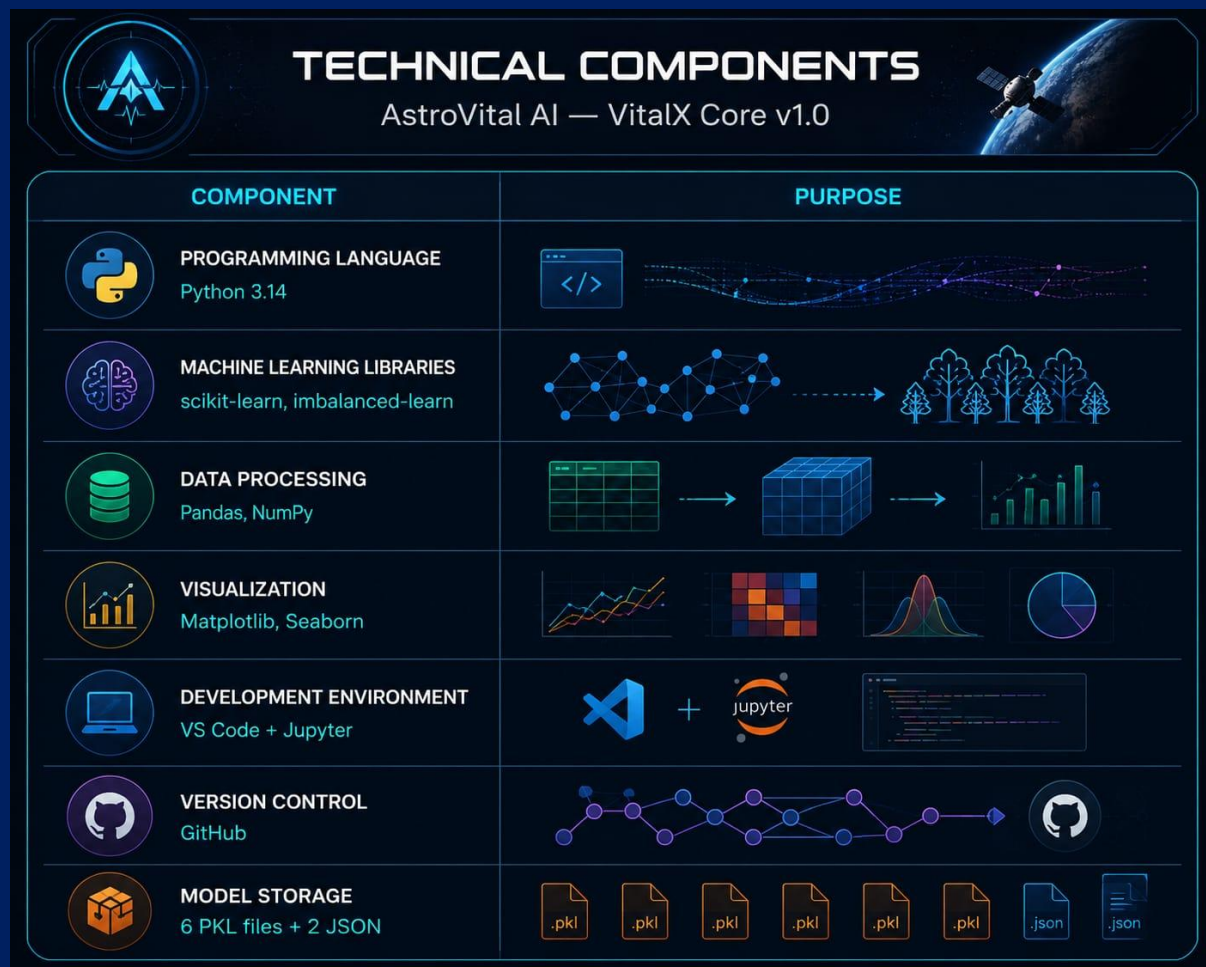
The Artemis II crew simulation yielded the following classifications under the optimized threshold of 0.27: Commander Reid Wiseman was classified as YELLOW, indicating a nominal physiological state consistent with mission-ready health. Mission Specialist Jeremy Hansen was similarly classified as YELLOW. Mission Specialist Christina Koch received a YELLOW classification, triggering an advisory status requiring enhanced monitoring and precautionary medical review. Pilot Victor Glover received a RED classification, indicating a critical health event threshold breach demanding immediate clinical attention and potential mission protocol adjustment.

ARTEMIS II — CREW HEALTH STATUS — DAY 06	
Reid Wiseman Commander	
● YELLOW - HOLD — Enhanced monitoring required	
RED P: 17.0% Threshold: 27%	
Victor Glover Pilot	
● RED - ABORT — Immediate CMO assessment	
RED P: 42.5% Threshold: 27%	
Christina Koch Mission Specialist	
● YELLOW - HOLD — Enhanced monitoring required	
RED P: 16.0% Threshold: 27%	
Jeremy Hansen Mission Specialist	
● YELLOW - HOLD — Enhanced monitoring required	
RED P: 22.0% Threshold: 27%	
⚠ CMO OVERRIDE AUTHORITY ACTIVE	



This simulation demonstrates several operationally critical system capabilities: the ability to simultaneously assess multiple crew members, the differentiation of health status across a realistic range of physiological conditions, and the generation of actionable, tiered alert outputs directly applicable to mission medical protocols. The crew health dashboard produced by this simulation — rendered through Notebook 9, Block 12 — provides a visual representation of the system's output format as it would appear to a crew medical officer in an operational deployment scenario. While the inputs to this simulation are synthetic, the pipeline, the models, the thresholds, and the decision logic are identical to those that would be applied to real sensor data in a deployed system.

SECTION K — TECHNOLOGY STACK:



VITALX CORE V1 was developed entirely within an open-source Python ecosystem, reflecting a deliberate commitment to reproducibility, accessibility, and the elimination of proprietary software dependencies that would constrain future deployment or collaborative extension of the system. The primary development language is Python 3.14, ensuring compatibility with the latest stable language features and library ecosystems at the time of project completion.

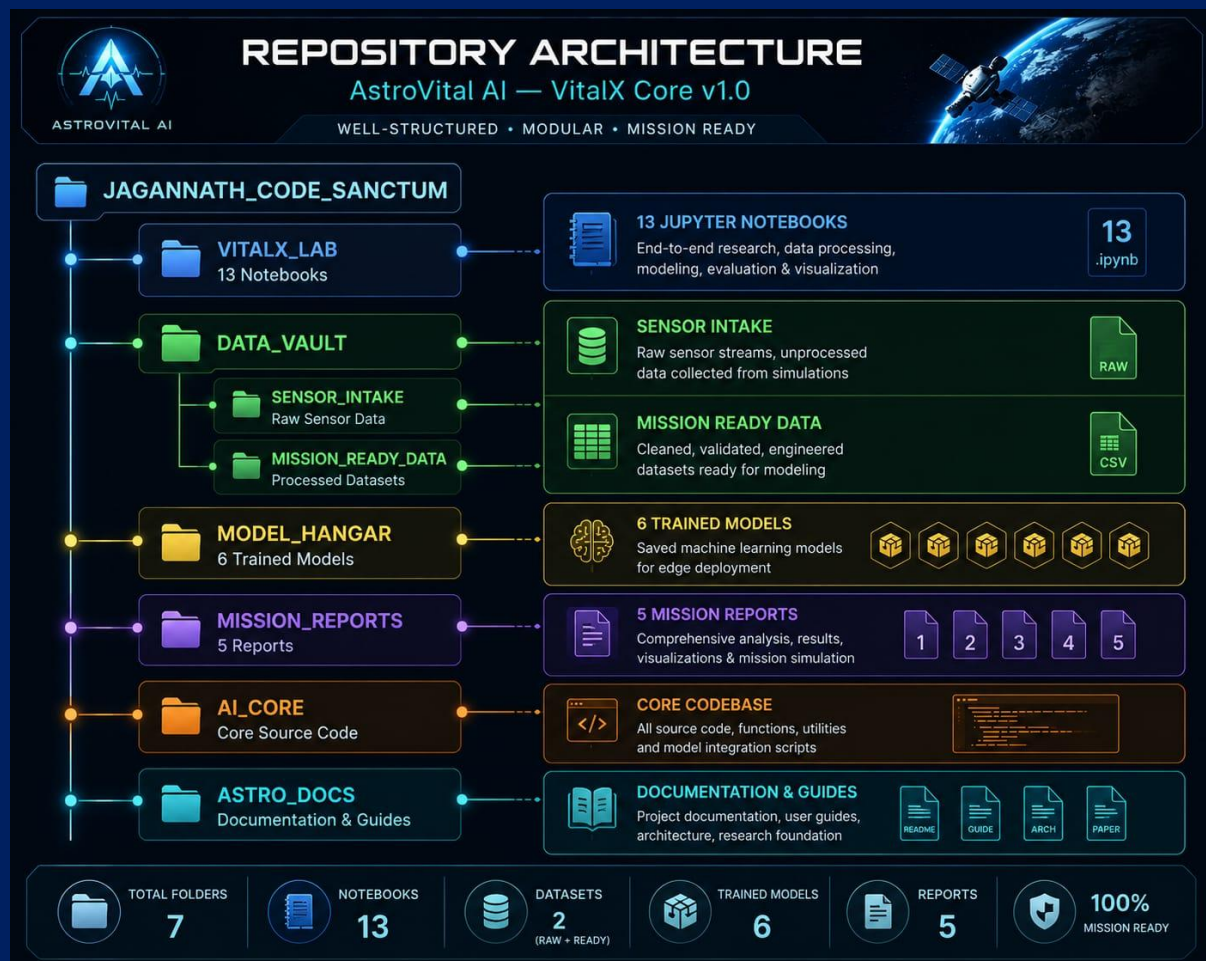
The machine learning layer of the system is built upon scikit-learn — the industry-standard Python library for classical supervised and unsupervised learning — supplemented by imbalanced-learn

for SMOTE-based class balancing. Data ingestion, manipulation, and pipeline management are handled through Pandas and NumPy, providing the full suite of array computation and tabular data processing capabilities required by the system's multi-stage pipeline. Visualization and performance reporting are produced using Matplotlib and Seaborn, enabling publication-quality graphical outputs for all model evaluation and biomarker analysis results.

Development was conducted within a Visual Studio Code environment augmented by Jupyter Notebook, providing an interactive, cell-by-cell execution environment suited to the iterative nature of machine learning research and the requirement for inline visualization of intermediate results. Version control and public archival of all project assets are managed through GitHub. The trained model artifacts — comprising six serialized model files in PKL format and two configuration files in JSON format — represent the deployable outputs of the system, encapsulating all trained parameters required for inference without retraining. This technology stack was selected not only for its technical capability but for its alignment with the open science principles that underpin the project's research philosophy.

SECTION L — REPOSITORY ARCHITECTURE:

The complete VITALX CORE V1 project is organized within a structured repository architecture designated JAGANNATH_CODE_SANCTUM — a naming convention that reflects the project's identity under the ASTROVITAL ecosystem and the Jagannath Command Centre organizational framework. The repository is designed for full reproducibility: every stage of the research pipeline, from raw data generation through model training, evaluation, and mission simulation, is traceable to a specific notebook and data file within the defined directory structure.



The primary research environment, VITALX_LAB, contains thirteen Jupyter Notebooks, each corresponding to a discrete

phase of the project's development — from dataset construction and exploratory analysis through innovation-specific development notebooks to the final integrated pipeline demonstration. The DATA_VAULT directory is partitioned into two subdirectories: SENSOR_INTAKE, which holds the raw and preprocessed dataset files, and MISSION_READY_DATA, which contains the final, biomarker-labeled, model-ready dataset versions. The MODEL_HANGAR stores the six serialized trained model files representing the deployable artifacts of all three innovation stages. MISSION_REPORTS contains five formal research and documentation reports generated throughout the project lifecycle, and the AI_CORE and ASTRO_DOCS directories house system configuration files and project documentation respectively.

This architecture was designed with two principles in mind: logical separation of concerns, ensuring that data, models, code, and documentation occupy clearly delineated spaces within the repository; and operational traceability, ensuring that any result presented in any report or visualization can be independently reproduced by navigating directly to the corresponding notebook and data source. The repository structure thus functions not merely as a storage system but as a transparent, auditable record of the entire research process.

SECTION M — HONEST LIMITATIONS:

Scientific integrity demands that the boundaries of any research system's validity be stated as clearly as its achievements. VITALX CORE V1 was developed under a set of constraints that meaningfully bound the scope of conclusions that can be drawn from its current performance metrics, and these constraints are documented here without reservation.

The most fundamental limitation is the synthetic nature of the training and evaluation dataset. While every parameter in the dataset was designed and validated against peer-reviewed physiological literature, synthetic data cannot fully replicate the complexity, inter-individual variability, temporal dynamics, and anomalous patterns present in real astronaut physiological recordings. Model performance metrics reported in this documentation reflect the system's behavior on data that, by construction, conforms to known statistical patterns. Performance on real mission data may differ — potentially substantially — in ways that cannot be fully anticipated from synthetic evaluation alone. Access to real astronaut physiological data remains the single most important prerequisite for transitioning VITALX CORE V1 from a research-validated prototype to a clinically deployable system.

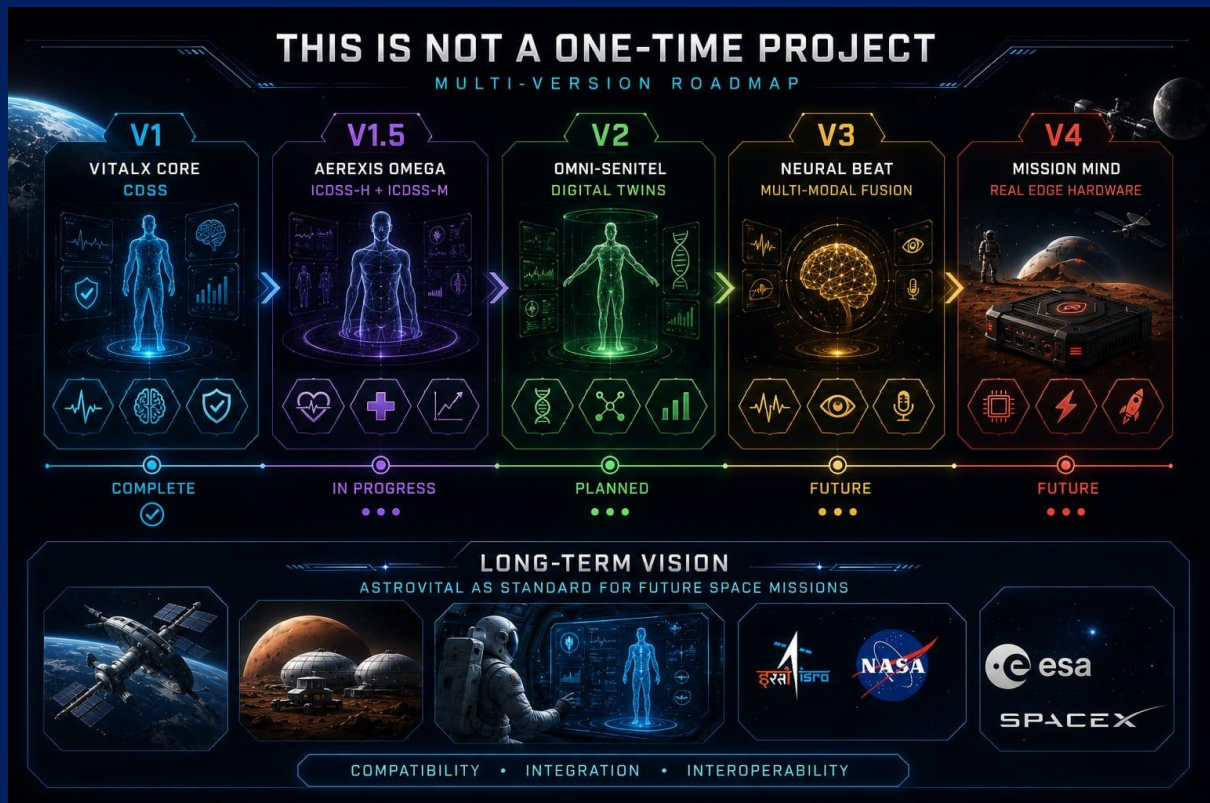
At the model level, RED class precision currently stands at 56%, indicating that approximately 44% of RED-class alerts generated by the system at the optimized threshold would be false positives in the current evaluation environment. While this tradeoff is deliberately accepted in favor of high recall — given the asymmetric consequences of missing a genuine critical event — it is acknowledged as a limitation that would require careful management in operational deployment, potentially through a two-stage confirmation protocol or adaptive threshold adjustment. The

sensor drift correction improvement of 9.23% MAE reduction, while statistically meaningful, reflects performance on radiation and artifact profiles estimated from literature rather than empirically measured in a space environment. The V1.5 & V2 roadmap iteration will address these limitations through the introduction of deep learning approaches — specifically LSTM networks — capable of modeling the temporal dynamics of physiological deterioration more accurately than the current static feature-based framework.

SECTION N — MULTI-VERSION ROADMAP:

VITALX CORE V1 represents the first completed milestone in a structured, multi-version research and development roadmap designed to progressively advance the ASTROVITAL system from its current foundation toward full operational mission readiness. Each version in this roadmap addresses a distinct and defined set of technical and clinical limitations inherited from its predecessor, ensuring that the development trajectory is driven by validated evidence of need rather than speculative feature expansion.

V1 — VITALX Core — is now complete, delivering a functional edge-deployed CDSS with three validated innovations, full explainability architecture, and demonstrated performance on an Artemis II simulation scenario. V1.5 — AEREXIS OMEGA — constitutes the next development milestone, extending the system into an Integrated CDSS-H (human health) and CDSS-M (mechanical systems) framework, fusing pilot biometric data with aircraft and spacecraft mechanical telemetry for a dual-domain anomaly detection and decision support capability. V2 — OMNISENSE — will introduce DIGITAL TWIN , deep learning architectures, specifically LSTM and Transformer models, capable of modeling the temporal evolution of physiological states and predicting deterioration trajectories rather than classifying instantaneous snapshots. V3 — NEURAL BEAT — will achieve multi-modal sensor fusion, integrating physiological, mechanical, and environmental sensor streams into a unified situational awareness model. V4 — MISSIONMIND — represents the terminal milestone of the current roadmap: full real-hardware edge deployment on mission-grade computational platforms, validated under conditions representative of actual spaceflight operations.



This roadmap reflects a research philosophy grounded in iterative, evidence-driven development. Each version builds upon the validated outputs of its predecessor, incorporating lessons learned, addressing documented limitations, and expanding capability in a direction defined by the operational requirements of human deep space exploration. The ASTROVITAL ecosystem, of which VITALX CORE V1 is the founding element, is designed to mature alongside the evolution of human spaceflight itself.

SECTION O — RESEARCH CONTRIBUTIONS:

VITALX CORE V1 makes several distinct and documentable contributions to the scientific literature in space medicine informatics and edge-deployed aerospace artificial intelligence. These contributions are presented not as incremental refinements of existing methods but as responses to specific, documented gaps in the peer-reviewed record.

The most significant contribution is the first known functional, edge-deployed Clinical Decision Support System designed explicitly for human spaceflight physiological monitoring, developed and validated in 2026 — twelve years after the research community first identified the need for such a system. By demonstrating that a three-class, real-time health status classifier can operate with competitive accuracy and clinically meaningful RED-class recall entirely without cloud dependency, this work provides a concrete existence proof for a class of systems whose feasibility had previously remained theoretical. The threshold tuning methodology developed for Innovation 1 — optimizing the classification probability threshold for medical minority class recall in a safety-critical context — represents a transferable contribution applicable to any imbalanced classification problem where the cost of false negatives vastly exceeds the cost of false positives. The parallel Random Forest and Decision Tree explainability architecture provides a reusable framework for deploying ensemble classifiers in contexts where transparency and auditability are non-negotiable operational requirements. The application of SMOTE to space medical synthetic data, and the demonstration of its efficacy in recovering RED-class recall, contributes a validated methodological precedent for future researchers working with similarly imbalanced aerospace health datasets. Finally, the data-driven biomarker ranking methodology of Innovation 6 provides the first evidence-based, space-

physiological-data-derived feature importance profile for a multi-domain astronaut health monitoring system. The primary publication targets for this work are *Acta Astronautica* and *npj Microgravity*, both of which represent appropriate venues for the intersection of aerospace engineering and space medicine research addressed by this project.

SECTION P — INTELLECTUAL PROPERTY AND COPYRIGHT:

All intellectual property contained within VITALX CORE V1 — including but not limited to the system architecture, algorithmic innovations, dataset design methodology, trained model artifacts, research documentation, visual design language, project nomenclature, and all associated written materials — is the original, sole, and exclusive creation of Gourgopal Mohapatra, developed independently without institutional, collaborative, or externally funded contribution of any kind.

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